FEDERICO GHIOLDI

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SUMMARY

- **Assistant Professor** at the Aerospace Dept. in Politecnico di Milano, specialized in reactive flow modeling and HPC
- **Research focus**: development of physics-based CFD methods for multiscale combustion in aerospace propulsion systems
- **Application domains**: hydrogen and ammonia combustion, high-speed and high-enthalpy propulsion systems in aerospace
- **European Projects Participant**: software developer in Horizon-2020 projects under Principal Investigator Prof. F. Piscaglia
- **Recognition**: "Key Innovator" by the European Commission Innovation Radar for contributions to GPU-accelerated reactive CFD

EDUCATION

Doctoral Degree in Aerospace Engineering **with honors**
Politecnico di Milano (Italy) *11/2019-01/2023*

Master Degree in Aeronautical Engineering **110/110 with honors**
Politecnico di Milano (Italy) *10/2016-04/2019*

TEACHING AND MENTORING

- Teacher (Ph.D. course) "*Bridging Experiments, High-fidelity Simulations, and Advanced Diagnostics in Propulsion Systems*" 04/2026 - present
- Teaching Assistant (M.Sc. course) "*Computational Techniques for Thermochemical Propulsion*" 09/2020 - present
- Teaching Assistant (B.Sc. course) "*Aerospace Propulsion*" 09/2024 - present
- Teaching Assistant (M.Sc. course) "*High Performance Computing in Aerospace*" 09/2025 - present
- Teaching Assistant (M.Sc. course) "*Aerodynamics*" 09/2022 - present
- Mentoring 3-5 students per year in their M.Sc. Thesis activities 09/2020 - present

RESEARCH VISITS

- Visiting Professor, Combustion, Acoustics & Flow Physics Lab. (CAPS) in ETH Zurich; Sup.: Prof. C. Frouzakis 02/2025 - 05/2025

RELEVANT PROJECTS

exaFOAM **05/2021 - 04/2024**

EU project focused on overcoming scalability limitations of reactive CFD simulations through heterogeneous HPC architectures within OpenFOAM. Main scientific contributions include:

- Designed and implemented a hybrid CPU–GPU framework for large-scale reactive flow simulations;
- Reformulated chemical ODE integration algorithms for efficient GPU-based massive parallelism;
- Developed a GPU-accelerated ray-tracing solver for radiative heat transfer in participating media.

□ *Innovations recognized as "Tech Ready" by the EU Commission Innovation Radar (12/2024) [LINK].*

ENGIMMONIA **09/2021 - 09/2024**

EU project targeting the development of ammonia-based propulsion systems for maritime decarbonization. Main scientific contributions include:

- Developed advanced injection and spray modeling strategies for dual-fuel (diesel–ammonia) combustion;
- Formulated a unified breakup modeling framework combining Huh-Gosman, KHRT, and Rayleigh–Taylor instabilities;
- Designed automated mesh generation strategies for deforming multi-block structured grids enabling robust mesh motion.

Green Propulsion Optimization at Vitesco **11/2019 - 01/2026**

Industrial research project focused on computational design methodologies for thermal management in electrified propulsion systems. Main scientific contributions include:

- Developed computational frameworks for the optimization of anisotropic multilayer thermal systems;
- Introduced reduced-order strategies to accelerate design-space exploration under multiphysics constraints.

TECHNICAL SKILLS

- **CFD development and analysis:** +7 years experience (OpenFOAM), mainly covering compressible flows, chemically reactive flow problems, combustion models, aerospace and hydrogen combustion, lagrangian particle tracking, HPC methods.
- **Programming capability:** +7 years experience in C++, C, CUDA-C, Bash, Python, and Matlab.
- **Other software worth of mention:** Fluent, Git, XFLR5, FreeFEM++, Office suite, LaTeX

PERSONAL

Languages **Italian:** mother tongue (native). **English:** professional proficiency (C1, TOEIC score 965). **Spanish:** elementary.
Attributes Competitive attitude; dynamic, creative, adaptive, results-oriented problem solver. Thrive in a team environment.
Speaking Confident and professional speaker; persuasive speaker in public, to groups, or via electronic media.
Voluntary work Volunteer experiences; help to elderly and people with disabilities. Member of Italian blood donation association.

JOURNAL ARTICLES

- F. Ghioldi, C.E. Frouzakis, F. Piscaglia, "*Differential Diffusion Effects on Chemical Mode Structure in Lean Premixed Hydrogen Flames via Computational Singular Perturbation analysis*", Physics of Fluids (2026), DOI
- F. Ghioldi, F. Piscaglia, "*Accelerated combustion simulations of hydrogen–air flames in the HYLON burner*", Physics of Fluids (2025), DOI
- F. Ghioldi, F. Piscaglia, "*A hybrid CPU-GPU Paradigm to Accelerate Reactive CFD Simulations*", International Journal for Numerical Methods in Fluids (2024), DOI
- F. Piscaglia, F. Ghioldi, "*GPU acceleration of CFD Simulations in OpenFOAM*", Aerospace (2023), DOI
- F. Ghioldi, F. Piscaglia, "*Acceleration of Supersonic/Hypersonic Reactive CFD Simulations via Heterogeneous CPU-GPU Supercomputing*", Computer & Fluids (2023), DOI
- F. Ghioldi, J. Hélie, F. Piscaglia, "*Multivariable Optimization of Pyramidal Compound Substrates for Cooling of Power-Electronics in Modern Hybrid and Electric Propulsion Systems*", Applied Thermal Engineering (2023), DOI
- F. Ghioldi, J. Hélie, F. Piscaglia, "*A Fast Computational Method for the Optimal Thermal Design of Anisotropic Multilayer Structures with Discrete Heat Sources for Electrified Propulsion Systems*", Int. J. of Heat and Mass Transfer (2021), DOI

JOURNAL ARTICLES UNDER REVIEW

- F. Ghioldi, C.E. Frouzakis, F. Piscaglia, "*Investigating combustion dynamics in hydrogen-air Rotating Detonation Engines using Computational Singular Perturbation*", Combustion and Flame (2026)
- F. Ghioldi, C.E. Frouzakis, F. Piscaglia, "*Mixing-state dependence of chemical explosivity and NO_x formation in hydrogen-air rotating detonation engines*", Combustion and Flame (2026)
- F. Piscaglia, S. Scoz, F. Lentini, F. Ghioldi, I. Spisso, J. E. Guerrero Rivas, S. M. Aithal, "*An implicit, density-based solver with novel Scalar-Coupled matrix assembly for supersonic reactive flows in OpenFOAM*", Computer & Fluids (2026)
- F. Ghioldi, C.E. Frouzakis, F. Piscaglia, "*A Lagrangian Framework for Computational Singular Perturbation Analysis of Turbulent Flames in Large-Eddy Simulations*", Physics of Fluids (2026)

BOOK CHAPTERS

- F. Ghioldi, F. Piscaglia, "*Accelerated Combustion Simulations of Hydrogen-Air Flames in the HYLON Swirled Injector in OpenFOAM*", In: L. T. De Luca, R. M. Doherty (Eds.), Energetic Materials and Techniques: Advances in Chemical Propulsion and Power Generation, Chapter 23, 1st Edition, Wiley-VCH GmbH, (2026), ISBN: 3527355456

CONFERENCE PRESENTATIONS

- F. Ghioldi, S. Scoz, F. Lentini, F. Piscaglia, "*Implementation and Assessment of a Non-Unity Lewis Number and Thermal Diffusion Transport Model for Hydrogen Combustion within LES in OpenFOAM*", The 21th OpenFOAM Workshop (OFW21) (2026)
- F. Piscaglia, F. Ghioldi, F. Lentini, S. Scoz, S.M. Aithal, J.E. Guerrero Rivas, I. Spisso„ "*External Solver Modules for Memory-Efficient Implicit Coupled Simulations Across All Flow Speeds*", 3rd Italian OpenFOAM User Meeting (2026)
- F. Ghioldi, C.E. Frouzakis, F. Piscaglia, "*A Generalized Dynamic Thickened Flame Framework for Hydrogen Combustion in Aero-engines with CSP Diagnostics*", The 20th OpenFOAM Workshop (OFW20) (2025)
- F. Piscaglia, F. Ghioldi, F. Chiappari, F. Lentini, I. Spisso, J. E. Guerrero, "*An OpenFOAM Library for the Implicit Density-based Solution of High-Enthalpy Hypersonic Flows*", 11th European Conference for Aeronautics and AeroSpace Sciences (EUCASS) (2025)
- F. Ghioldi, F. Piscaglia, "*Investigating Reactive Flow Dynamics and Pollutant Formation in Aero-Engines via a Dynamic Thickened Flame LES Combustion Model in OpenFOAM*", 11th European Conference for Aeronautics and AeroSpace Sciences (EUCASS) (2025)
- F. Ghioldi, F. Piscaglia, "*Accelerating Hydrogen Combustion Simulations in Aero-Engines with Advanced CFD and GPGPU Techniques*", AIAA SciTech Forum (2025)
- F. Piscaglia, F. Ghioldi, J. E. Guerrero Rivas, I. Spisso, "*A Unified Implicit Density-based GPU Coupled Solver for Hypersonic Flows Computations in OpenFOAM*", AIAA SciTech Forum (2025)
- F. Ghioldi, J. E. Guerrero Rivas, I. Spisso, F. Piscaglia, "*A Unified GPGPU Implicit Density-Based Solver for Hypersonic and Combustion Simulations in OpenFOAM*", 2nd Italian OpenFOAM User Meeting (2024)
- F. Ghioldi, F. Piscaglia, S. M. Aithal, "*Efficient Design of Low-NOx Hydrogen Combustors by GPU-Accelerated Simulations in OpenFOAM*", 35th Int. Conf. on Parallel Computational Fluid Dynamics (2024)
- F. Piscaglia, E. Gallorini, F. Ghioldi, A. Marie, J. P. Fradin, J. Hélie, "*Experimental Validation of Thermal-Adjoint Topological Optimization for Cooling Plate Design*", 30th International Workshop on Thermal Investigation of ICs and Systems (2024)
- F. Piscaglia, F. Ghioldi, F. Magugliani, S. M. Aithal, "*Accelerated CFD of Aerospace Propulsion Systems in OpenFOAM*", 14th Int. Workshop on Combustion and Propulsion (2024)
- F. Ghioldi, F. Piscaglia, "*GPU Acceleration of CFD Simulations in OpenFOAM*", 18th OpenFOAM Workshop (2023)
- F. Ghioldi, F. Piscaglia, "*GPU-Accelerated Simulation of Supersonic Combustion in Scramjet Engines by OpenFOAM*", 33rd International Conference on Parallel Computational Fluid Dynamics (2022)
- D. Costero, F. Ghioldi, S. M. Aithal, F. Piscaglia, "*Novel Developments for Rapid Reactive CFD Simulations of Dual-Fuel IC Engines*", 33rd Int. Conf. on Parallel Computational Fluid Dynamics (2022)
- E. Gallorini, F. Ghioldi, S. M. Aithal, F. Magugliani, F. Piscaglia, "*A Methodology for the Aero-thermal Optimization of Hybrid and Electric Propulsion Systems*", 9th Annual OpenFOAM User Conference (2021)
- F. Ghioldi, E. Gallorini, S. M. Aithal and F. Piscaglia, "*A CFD Methodology for the Optimal Thermal Design of the Propulsion System in Electric Motors*", AIDAA, XXVI International Congress (2021)
- F. Ghioldi, F. Piscaglia, "*A CPU-GPU Paradigm to Accelerate Turbulent Combustion and Reactive-Flow CFD Simulations*", 8th Annual OpenFOAM User Conference (2020)